

Antecedents and Outcomes of Informal Learning Behaviors: a Meta-Analysis

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Abstract

Purpose Over the past two decades, research has shown a growing consensus that 70% to 90% of organizational learning occurs not through formal training but informally, on-the-job, and in an ongoing manner. Despite this emerging consensus, primary data on the nature and correlates of informal learning remains sparse. The purpose of this study was to provide an integrative definition of informal learning behaviors (ILBs) and to synthesize existing primary data through meta-analysis to explore ILB correlates.

Design/Methodology/Approach Given that there has been little systematic treatment of ILBs, we defined their construct domain and tested relationships suggested by our research questions with antecedents (personal factors, situational factors) and outcomes (attitudes, knowledge/skill acquisition, performance) using random effects meta-analyses ($k = 49$, $N = 55,514$).

Findings Our results showed both personal and situational antecedent factors to be predictive of ILBs, as well as ILB–outcome relationships.

Implications Findings indicate that engagement in ILBs for working adults is linked to valued criteria such as attitudes ($\rho = .29$), knowledge/skill acquisition ($\rho = .41$), and performance ($\rho = .42$). We provide suggestions for future research and actionable advice for organizations to support the development of ILBs.

Originality/Value Although hundreds of studies and over a dozen meta-analyses have explored the nature and effectiveness of formal learning in the workplace, our work is the first attempt to conceptualize a unified definition of ILBs and to aggregate primary data on ILB correlates using meta-analysis.

Keywords Performance · Incidental learning · Field-based learning · Training · Experiential learning

Employee learning has a profound impact on the success of modern organizations (Bartlett & Ghoshal, 1998). Formal training programs (i.e., curriculum-based offerings provided, sponsored, and/or supported by the employee's organization) are intended to promote such learning, and hundreds of primary studies show that well-designed training can boost learning and performance. The effectiveness of formal training programs has been well documented through meta-analytic research for both individuals and teams (Colquitt et al., 2000; Salas et al., 2008), for young and old learners (Callahan et al., 2003), for a variety of program types (Arthur et al., 2003), and for various program aims (Keith & Frese, 2008; Morris & Robie, 2001).

Formal training programs, however, cannot adequately prepare employees for all possible scenarios and are typically not designed to equip individuals for ongoing, on-the-job learning (Blume et al., 2010). Employees also need to acquire new knowledge and skill on an ongoing basis through informal, non-curricular means (Tannenbaum et al., 2013, 2010). In fact, many have suggested that the *majority* of learning and development in the workplace occurs through experience (Center for Workforce Development, 1998; McCauley &

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Brutus, 1998; Morrison et al., 1992). Estimates of the proportion of organizational learning that takes place informally range from 70% (Cseh et al., 2000) to 80% (Koopmans, Doombos, & Eekelen, 2006; Marsick & Watkins, 1990) to over 90% (Flynn et al., 2006; Tannenbaum, 1997). The simple fact that employees spend far more time “working” than in formal training provides a *prima facie* case for the potentially large impact of informal learning.

Despite the potential upside of informal learning behaviors (and in contrast to research that examines formal training), there exists no comprehensive theory or meta-analysis that has systematically explored their potential antecedents and outcomes. Noting the prevalence with which informal learning actually occurs, we believe a theoretical review and quantitative analysis of this domain are warranted. Thus, we have two main goals for this work. First, we define informal learning behaviors (ILBs) to specify the construct domain for our review and meta-analyses. In doing so, we examine categories of personal and situational *antecedents* that might encourage or deter engagement in ILBs in working adults. We also examine relationships between ILBs and different effectiveness *criteria*. Second, we conduct a meta-analysis to address research questions focusing on suggested antecedents and outcomes of engagement in ILBs. We conclude with an agenda for future research and provide actionable considerations for the advancement of ILBs in organizations.

It should be noted that although meta-analyses can have a “chilling” effect on subsequent research (Chan & Arvey, 2012), our intent is exactly the opposite. Instead, where theory is lacking, findings contradictory, or cumulation needed, meta-analysis provides an ideal solution (Schmidt & Hunter, 2001). Our intent is to drive and leverage the understanding and effectiveness of ILBs in organizations by providing a preliminary, integrative examination of ILBs that can be subsequently refined and contextualized. Below, we synthesize existing findings and highlight places where theory and empirical research can be further developed.

What Is Informal Learning and Informal Learning Behavior?

Terms such as “informal learning” and “formal learning” have been used ambiguously, often confounding gains in knowledge/skill (K/S) with the behaviors that result in such gains. To separate the two concepts, we propose a clearer delineation, one that focuses on behaviors, such that “formal learning” refers to K/S gained through *formal learning behaviors* (FLBs; e.g., attending a class, taking notes during a lecture or prescribed class activity, completing homework or assigned readings), whereas “informal learning” refers to K/S gained through ILBs (e.g., observing a coworker, discussing a problem with a supervisor, asking questions while on the job).

Thus, to be clear, we focus on informal learning behaviors, rather than on learning itself (although we explore that as well as an outcome of ILBs). To clarify what sets ILBs apart from FLBs (primarily to provide a point of comparison for our discussion), we can specify a definition for FLBs as follows:

Formal learning behaviors (FLBs) are curricular behaviors and activities pursued in service of knowledge and skill acquisition that take place within formally-designated learning contexts. Such activities are organizationally sanctioned, externally directed, and are classroom- and/or course-based. Individuals generally initiate and engage in FLBs that correspond with activities prescribed/scheduled by a syllabus or instructor rather than by their own curiosity. They often have well established learning objectives; typically proceed in a linear fashion; and have a discrete beginning and end.

Informal learning behaviors, by comparison, are qualitatively different than FLBs. We offer an integrative definition that is consistent with, but builds upon, the framework set out by Tannenbaum et al. (2010):

Informal learning behaviors (ILBs) are non-curricular behaviors and activities pursued in service of knowledge and skill acquisition that take place outside formally-designated learning contexts. Such activities are predominantly self-directed, intentional, and field-based. Informal learning behaviors are not syllabus-based, discrete, or linear.

Broadly speaking, informal learning behaviors are non-curricular and they are highly experiential; they occur in the workplace outside of formal learning contexts, via observing, asking questions, practice, and so forth (Sambrook, 2005). ILBs can be characterized by the absence of external instructions and formal structures directing what and how to learn. In other words, informal learning does not have a designated/assigned set of knowledge and skills to be mastered. Rather, ILBs are self-initiated, intrinsically driven, individually controlled, and aimed towards obtaining goals set by the learner rather than those established by a trainer, instructor, or organization (e.g., Noe, Tews, & McConnell-Dachner, 2010; Stamps, 1998).

Both the context (on the job, but not within a situation specifically designed and structured for learning) and the activities of the learner are key defining characteristics of informal learning (Marsick et al., 1999). The stipulation of a *non-training context* is important because the same behavior (e.g., asking questions) can occur in either a formal or a non-formal context. Given that behaviors are rarely independent of context (Ross & Nisbett, 1991), ILBs are identified as particular kinds of behavior within a particular context. Thus, it is not

merely the structure of the learning environment that matters but also the experience of the learner while engaging in these behaviors.

Intentionality is also an important aspect of informal learning and ILBs because organizational learning in general can be broadly classified by the degree to which it occurs purely by accident (unintentionally or incidentally) or through conscious deliberation (intentionally). Clearly, learning can occur implicitly, incidentally, or unconsciously (Ellinger, 2005; Gola, 2009; Marsick & Volpe, 1999). However, Noe et al. (2010) argue for a definition of informal learning in organizations that involves conscious action and doing (that is, it requires an agent rather than occurring passively). This is consistent with other areas of organizational behavior; for example, most credible theories of workplace motivation focus on conscious, goal-directed behavior to the exclusion of accidental or unconscious drives (Donovan, 2001). The difference then, between informal learning and incidental or implicit learning (and their associated behaviors), is the presence of intentional, conscious direction. As summarized by Tannenbaum et al. (2010):

“...the *intent to learn or improve* is what differentiates informal learning from incidental [unintentional] learning...the learning process has to be driven by the individual” (p. 306, emphasis added).

Given the nature of the field, and the observation that various terms have been used to refer to a single concept (Cseh et al., 2000), it makes sense to explicitly state where informal learning is related to, but distinct from, other existing constructs.¹ We acknowledge that there are similarities to incidental/implicit learning, in that both occur in organizations and are not formally directed; however, incidental, unconscious, implicit, and accidental learning lack the critical conscious, goal-directed nature of ILBs. Both ILBs and continuous learning are characterized by a lack of programmatic approach (LeClus, 2011); however, the latter is usually used to refer to formal courses or professional development opportunities that are simply pursued at the individual’s leisure, which are still formal learning opportunities. Mentoring and coaching arrangements have elements of informal learning, in that they often take place outside a formal context (Eraut, 2004); however, unlike ILBs, there is usually an agreed-upon outcome/criteria and linear progression for such efforts. Social, vicarious, and network learning differ from ILBs in that they require some interaction with (or observation of) other individuals (Gola, 2009). There are many more related terms, and our list could be longer, but this should provide a general overview of how we have bounded ILBs. In sum, compared to related constructs, informal learning behaviors of working adults are self-guided, intentional, and occur

independently from formally directed learning requirements and context. In practice, ILBs proceed in a non-linear fashion, with a less-clearly set beginning or end, with varying levels of intensity over time, and in a variety of job contexts.

Given that ILBs are key to informal learning, a conceptual framework that organizes the nomological network surrounding ILBs is needed. It is not our intent to explore every possible construct and correlate that could exist in the greater nomological network surrounding ILBs. In an effort to clearly bound the ILB construct and to guide the domain of our meta-analyses, we restrict ILBs to conscious engagement occurring outside of formal learning programs. Towards that goal, we provide a more general framework that develops a common understanding of the construct, fosters practical recommendations, and provides a starting point for future research. Below, we expand upon the Tannenbaum et al. (2010) conceptions and further develop an ILB framework. In the process, we supplement it with conceptual work from the field to generate research questions about both antecedents and outcomes of engaging in ILBs. In each case, we derive the questions as logical exploratory follow-ons to findings in the literature. That is, because of the still-developing nature of IL and ILB constructs, we felt it better to pose research questions that, although consonant with the extant IL literature, are more tentative than fully developed hypotheses.

Antecedents of Informal Learning Behaviors

Tannenbaum et al. (2010) suggest very broadly that two antecedent categories of ILBs must be considered: *personal* and *situational*. Given the broad nature of such a framework, we have attempted to add some detail and granularity to their work. Table 1 conceptualizes antecedents of ILBs at three levels of granularity, beginning at the most general level (as articulated by Tannenbaum et al.), labeled Construct Specificity Level 1 (i.e., Personal Antecedents, Situational Antecedents, Outcomes). We then proceed with greater detail of construct specificity at level 2 (e.g., Individual Predispositions, Support) and at level 3 (e.g., Learning Goal Orientation, Formal Rewards for Learning). In the process, we generate research questions for meta-analyses to be conducted at the three levels of specificity.

Personal Antecedents

Based upon Tannenbaum et al. (2010), previous theoretical reviews (e.g., Noe et al., 2010), and upon our examination of variables most often studied in informal learning research, we consider two clusters of personal antecedents for this meta-analysis: (1) individual predispositions and (2) demographics.

Individual Predispositions By individual predispositions, we cluster together individual traits or trait-like orientations that

¹ We thank an anonymous reviewer for helping us to hone this point.

Table 1 Theoretical antecedents and outcomes of informal learning at three levels of construct specificity

Construct Specificity Level 1	Construct Specificity Level 2	Construct Specificity Level 3
Personal antecedents	Individual predispositions	• Personality/propensity factors (e.g., conscientiousness, curiosity, adaptability) • General learning-related motives (e.g., perceived need for informal learning, learning goal orientation)
	Demographics	• Age, Education, Sex, Income, Rank/Tenure, Experience, Marital Status
Situational antecedents	Job/task characteristics	• Demands • Resources • Control/autonomy
	Support	• People support (e.g., supervisors, peers, role models, partners) • Formal organizational support (e.g., rewards, processes, systems) • Informal organizational support (e.g., climate, social capital, culture, norms, perceived organizational support)
	Opportunities for learning	• Potential for new learning • Lower workload • Time
Outcomes	Attitudes (i.e., positive general work attitudes [e.g., job satisfaction, organizational commitment, perceived job insecurity, engagement])	
	Knowledge/skill acquisition (e.g., proficiency, knowledge acquisition, job content learning, built knowledge base, core skills)	
	Performance (e.g., effectiveness, salary, promotions, project performance)	

Construct Specificity Level 1 reflects the highest-order (broadest) constructs, whereas Construct Specificity Level 3 represents the lowest-order (most specific) constructs

predispose individuals to engage in learning-oriented behaviors. These would include certain *personality/propensity* factors, such as those assessed by the Big Five variables. Noe, Tews, and Marand (2013) have found significant links between informal learning and the Big Five, with extraversion, openness to experience, and agreeableness exhibiting the highest positive correlations. Whereas a case can certainly be made that emotional stability and conscientiousness are positively related to seeking opportunities for development (see Simmering, Colquitt, Noe, & Porter, 2003), the bundle of extraversion, openness, and agreeableness seems to empirically describe individuals who will willingly look for and embrace opportunities for learning (Noe, et al., 2013). The predispositions category also includes other semi-stable, trait-like individual differences such as adaptability (Mumford, Baughman, Threlfall, Uhlman, & Costanza, 1993) and curiosity (Reio & Wiswell, 2000).

Beyond personality/propensity factors, some researchers have also proposed that stable *general learning-related motives* such as learning goal orientation, motivation to learn, and generalized self-efficacy would impact individuals' engagement in informal learning (Choi & Jacobs, 2011). Learning goal orientation is defined as an individual's intention to engage in challenging activities, an eagerness to improve oneself, and a tendency to use one's past performance as a standard for evaluating current performance (Button, Mathieu, & Zajac, 1996). Previous research is consistent with the assumption that individuals with

this type of motivation would naturally be inclined to engage in self-development activities, including informal learning (Boyce et al., 2010; Brett & VandeWalle, 1999; Choi & Jacobs, 2011; Klein, Noe, & Wang, 2006; Orvis & Leffler, 2011). Research has also found significant correlations among individuals' motivation to learn (Noe & Schmitt, 1986), informal learning, and related constructs (Choi & Jacobs, 2011; Lohman, 2005; Moon & Na, 2009). Self-efficacy, defined as an individual's beliefs about their capabilities to organize and execute courses of action required to achieve certain levels of proficiency or performance (Bandura, 1994), has been related positively to intentions to participate in self-development activities (Maurer & Palmer, 1999; Molloy & Noe, 2010; Noe & Wilk, 1993; Orvis & Leffler, 2011) and informal learning (Noe et al., 2013).

Collectively, individuals' predispositions represent potentially powerful drivers towards engaging in learning behavior. In fact, Choi and Jacobs (2011) found that a distillation of learning orientation, self-efficacy, and motivation to learn was a strong predictor of not only informal learning but also motivation to engage in a formal training program designed to facilitate informal learning. Given the importance of these factors, it seems logical to ask:

Research question 1: Are individuals' predispositions, both personality/ propensity factors and general learning-related motives, related positively to engagement in ILBs?

Demographics Demographics (e.g., age, education, sex, income, rank/tenure, marital status) have often been measured alongside learning-related outcomes, such as informal learning. However, rarely have participants' demographics been treated as substantive predictors of ILBs; they are typically considered as simply potential covariates (Berg & Chyung, 2008). Therefore, the relevance of employees' demographics to ILBs remains unclear. Accordingly, we pose the following question:

Research question 2: Are common demographic factors related to individuals' engagement in ILBs?

Situational Antecedents

As noted by Tannenbaum et al. (2010), the characteristics of the situation or domain may also influence the degree to which informal learning behaviors occur. Although there are several ways to categorize situational antecedents of ILBs (e.g., Noe et al., 2010; Tannenbaum et al., 2010), we suggest that in the interest of parsimony, these factors can be grouped into three clusters: (1) job/task characteristics, (2) support, and (3) opportunities for learning.

Job/Task Characteristics From a practical standpoint, one important task characteristic is whether individuals have sufficient *resources* to engage in elective ILB activities. Mathieu et al. (1992) illustrated how situational task constraints (e.g., inadequate equipment and supplies, insufficient authority to complete tasks) can undermine motivation to learn. Similarly, LePine et al. (2004) found that stress associated with hindrances in the learning environment reduces motivation to learn (and learning performance).

Of course, individuals are less likely to initiate ILBs if they feel they lack the authority to do so. Previous research suggests that the degree to which individuals perceive *control/autonomy* over their work has a strong impact on the extent to which they engage in discretionary organizational citizenship-type behaviors such as teamwork and participation in non-mandated formal training. For example, Gijbels et al. (2012) and Ouweneel et al. (2009) demonstrated that when individuals have control over their work, ILBs increase. Further, greater control or task autonomy may allow individuals to create learning opportunities and a malleable environment may enable a proactive employee to generate learning opportunities. In accordance with these indications from the literature, we pose the question:

Research question 3: Are job/task characteristics associated with ILBs, such that enabling factors (e.g., resources, control/autonomy) foster ILBs while constraining factors (e.g., demands) inhibit ILBs?

Support Because employees are more likely to engage in elective behaviors when they are reinforced to do so (LePine et al., 2002) and ILBs are an elective form of behavior, various sources of support should bolster ILBs. Support may be derived from three different sources. First, support may come from different *people*, such as supervisors or coworkers (Choi & Jacobs, 2011; Maurer & Tarulli, 1994; van der Heijden et al., 2009). For instance, supportive supervisors often encourage employees to engage in activities that lead to professional development or career advancement (Rooney & Gottlieb, 2007). Moreover, they provide employees with decisional discretion and encourage them to solve their own problems (Rooney & Gottlieb, 2007; Yukl, 2010). It is likely that employees who feel supported by their supervisors and coworkers develop a heightened sense of security and self-confidence to try new things.

A second source of support may come in the form of *formal organizational* programs such as incentives, practices, policies, and rules (Rowden, 2002; Rowden & Conine, 2005). Incentives can provide a strong, formal, tangible support for informal learning. For example, Rowden (2002) and Rowden and Conine (2005) found positive correlations between recognition programs and informal learning. Rowden and Conine (2005) and Moon and Na (2009) found that compensation or merit supportive of informal learning promoted informal learning activities. Skule (2004) included among the determinants of “learning intensive jobs” the rewarding of proficiency through higher wages.

Finally, there may be *informal organizational supports* as reflected in employees' attitudes and beliefs about the support they receive from their organization (Rowden & Conine, 2005; Tannenbaum et al., 2010). These could include variables such as an organizational learning climate, discussions of learning opportunities associated with assignments or projects, the use of team debriefs that could encourage reflection behaviors, and leaders who “think aloud” about their experiences to encourage others to reflect on their own. In line with this, Rowden and Conine (2005) found empirical support for the positive relationship between perceived organizational climate and informal learning. Likewise, Berg and Chyung (2008) found a positive relationship between learning culture and informal learning, and Jeon and Kim (2012) found a positive relationship between “innovative culture” and informal learning. Given the research on the effects of support, we ask:

Research question 4: Are perceptions of support (people, formal organizational, informal organizational) associated positively with ILBs?

Opportunities for Learning Individuals should be more likely to engage in ILBs when ample opportunities exist (Noe et al., 2010). Opportunities for learning refer to the extent to which the situation affords potential for learning. Availability of feedback

can be seen as one component of learning environment “richness,” because employees can take advantage of feedback to understand how to guide their future behavior (de Groot et al. 2012; Kluger & DeNisi, 1996). In some instances, potential for learning may be inherent in the environment. For example, Molloy and Noe (2010) linked organization type or sector (e.g., government, military, civilian), and whether the organization was in a state of growth or decline, with the likelihood that individuals would engage in informal learning behaviors. This category might also include factors such as organizational structure (less structured environments are thought to permit exploration and alternative ways to conduct elective activities such as ILBs; Moon & Na, 2009). Further, Klein et al. (2006) suggested that when individuals lack the time needed to participate in learning opportunities, they will be less likely to engage in them because they are less motivated to do so, whereas when time is more plentiful (or there is a lighter workload), individuals have availability to do so. Accordingly, we ask:

Research question 5: Are available opportunities for learning (i.e., potential for learning, lower workload, time) positively associated with engagement in ILBs?

Outcomes of Informal Learning Behaviors

An implicit assumption throughout our discussion so far is that engagement in ILBs should pay dividends in terms of valued outcomes. Although explored in different areas, no systematic review or meta-analysis has linked ILBs with desired outcomes across studies and contexts. Accordingly, we consider three common criteria and explore their links with ILBs: (1) attitudes, (2) knowledge/skill acquisition, and (3) performance.

Attitudes refer to the summary cognitive and affective evaluation of one’s organization, job, or task. These broad evaluations of the work and surrounding environment are important because they are associated with performance on formally assigned tasks. Perhaps more importantly, positive general work attitudes drive engagement in critical voluntary behaviors such as teamwork and various types of contextual performance. Based on related research (e.g., Meyer et al., 2002), this might include constructs such as contentment with work, perceived job security, degree of work involvement, and organizational commitment. For example, Gonzales (1985) reported a positive correlation between employees’ informal learning and their degree of work involvement. More recently, Richter et al. (2011) found a significant positive correlation between employees’ informal learning activities and their work engagement. Accordingly, we propose the question:

Research question 6: Is engaging in ILBs associated positively with individuals’ work-related attitudes?

A second criterion commonly assessed in any learning context is the acquisition of knowledge and skill (Alliger et al., 1997). Individuals who constantly engage in trial-and-error tactics and feedback seeking are more likely to learn (Marsick et al., 1999). As expected, several previous studies have found somewhat robust links between engagement in ILBs and the acquisition of both knowledge (Matsuo & Nakahara, 2013) and skill (Spreitzer, McCall & Mahoney, 1997). Because of the conceptual link between ILBs and knowledge/skill gain, we ask:

Research question 7: Is engaging in ILBs associated positively with knowledge/skill acquisition?

Finally, beyond enhanced work attitudes and knowledge/skill, we anticipate that employees who engage in informal learning activities are likely to be higher performers. Research shows that experts are higher performers as a result of years of self-directed behaviors, punctuated by trial-and-error and experiential learning of cue-outcome relationships (Ericsson & Charness, 1994). Further, research has also demonstrated that when learners actively engage a problem and navigate their way towards a solution, they end up with better performance outcomes in the long run (Bell & Kozlowski, 2008). These findings have been mirrored in the research on informal learning outcomes. For example, Spreitzer et al. (1997) observed a moderate positive correlation between individuals’ levels of informal learning and job performance. Similarly, Wasiyo (2010) found a positive correlation between project managers’ engagement in informal learning and their effectiveness. Given these findings, we ask:

Research question 8: Is engaging in ILBs associated positively with performance?

We next describe a series of meta-analyses of the literature that examine the proposed research questions.

Method

Study Search

Using a number of databases, we searched for published studies through the year 2015. Given the breadth in domains where informal learning-related research exists, we used psycINFO, ERIC, PubMed, Scopus, Google Scholar, and Dissertation Abstracts International to conduct a broad search across fields and disciplines. For search and coding purposes, informal learning behaviors were operationalized as self-directed, non-curricular activities outside of formal learning contexts pursued in service of knowledge and skill acquisition. Search terms used to locate primary studies were *informal learning*, *field-based learning*, *situated learning*, *self-directed learning*, and *on-the-*

job learning. We selected for inclusion studies that were rated by two of the authors as measuring constructs that conformed to our operationalization. This included activities such as self-experimentation, learning with others, learning from colleagues, information seeking, learning by doing, and study-defined informal learning activities.

To maintain the scope of our study, we excluded studies that did not fit our operationalization, including those measuring constructs such as experiential learning, debriefing, adventure learning, incidental learning, mentoring, socialization experiences, and error management learning. Samples were limited to high-school age and above, given our focus on working adult populations, and we did not include any non-English research. Otherwise, no specific populations, study designs, time periods, years, or geographical locations were rejected. Except where otherwise indicated, all analyses include both published and unpublished samples. Our search yielded 3146 results. Of these, we read through all abstracts to cast the widest net possible for any potentially relevant data. We identified and ordered for deeper review 428 sources where the abstract suggested any chance whatsoever of including an empirical assessment of informal learning.

Coding Data

A coding schema was set up in advance and every data point (18 variables² $\times$ 376 effect sizes = 6768 data points) was coded by two authors (experts in the field) to confirm a common frame of reference. Studies that did not include sufficient data to calculate effect sizes were excluded from the study, and no data were discarded as potential outliers. Initial coding discrepancies (204 discrepancies / 6768 = 3.01%) were resolved through joint discussion with a third author. With each coding decision, we followed standard procedures to avoid creating heterogeneous categories (Aguinis, Gottfredson & Wright, 2011), the “apples and oranges” problem of meta-analysis (Cortina, 2003).

All antecedents and outcomes were categorized in line with our discussion in the introduction and Table 1. Personal antecedents included individual predispositions such as personality/propensity factors (i.e., trait-like orientations conducive to engagement in ILBs, such as openness to experience) and learning-related motives (i.e., state-like orientations conducive to ILBs, such as motivation to learn). Personal antecedents also included common demographics (e.g., age, education, marital status, and experience).

Situational antecedents covered common job/task characteristics, including task/job demands, availability of critical resources, and perceptions of control/autonomy over one’s work.

Perceived support was reported in three ways: interpersonal support from other people (e.g., social support, support from peers), informal support (e.g., supportive learning environment, opportunities for feedback), and formal support (e.g., formal recognition systems, compensation tied to ILBs, merit systems). Learning opportunities referred to potential for learning (e.g., learning-rich tasks, interactive environments) and time available (or time constraints reverse-coded).

Regarding outcomes, general affective and cognitive evaluations of the workplace, organization, or particular job were coded as attitudes (e.g., job satisfaction, organizational commitment, engagement). We coded measures of knowledge/skill acquisition (e.g., proficiency, knowledge acquisition, job content learning, building core skills). We coded individual performance in line with the view of Guion (1998) as the individual process or outcome of goal-directed, organizationally relevant behavior (e.g., project performance, effectiveness, promotions, salary raises).

When there was relatively little known about a particular topic, or when primary data are aggregated from a wide variety of sources, it is imperative to explore the quality of the primary data, the potential for publication bias, and the potential influence of methodological rigor on the magnitude of observed study effect sizes (Borenstein et al., 2009). Given that our search cast a broad net and included findings from multiple outlets and disciplines, we also coded for additional methodological moderators related to publication quality. This consisted of several items that would enable us to assess the rigor of primary studies in four ways: (a) source impact factor, (b) citation count, (c) study design (1 = cross-sectional, 2 = lagged, 3 = longitudinal, or 4 = experimental), and (d) measurement source (1 = same, 0 = different). These were combined and tested in a robust multi-level model, taking into account year of publication as well.

Analysis

We employed the random effects meta-analysis methods and software of Hunter and Schmidt (2004), using reliability estimates available to create artifact distributions and correct effect size estimates for artifactual variance associated with measurement unreliability. We corrected for measurement unreliability in both predictor and criterion variables (where sufficient information was available) and report the mean of those distributions for each estimate. Note that corrections for both predictor and criterion artifacts lead to estimates that more closely approximate the “true” effect size than uncorrected coefficients. Where information was not available, effect sizes are conservative estimates because they do not apply any unreliability correction. We also corrected for sampling error. For gauging the heterogeneity of effect sizes, we followed Hunter and Schmidt’s recommendation concerning the use of 80% credibility intervals, percent variance accounted for by statistical artifacts (above

² Author, year, observed effect size, IV reliability, informal learning reliability, IV description, IV level 1, IV level 2, IV level 3, temporal precedence, design, source IV, source DV, measurement source difference, informal learning description, notes, source

75% would suggest no additional moderators), and 95% confidence intervals (CIs) to infer whether an observed effect differs from zero (see Whitener, 1990).

Although some studies reported more than one effect size of interest, each sample must only contribute one effect size estimate towards a given meta-analytic estimate to maintain the independence of observations assumption (cf., Bijmolt & Pieters, 2001; Borenstein, Hedges, Higgins & Rothstein, 2009). For example, the study by Berg and Chyung (2008) reported the correlation of age with nine different informal learning activities, yielding nine possible coefficients that could represent the sample. Although it is optimal to compute composite reliabilities and effect sizes in line with Mosier (1943), when necessary information is not available (e.g., intercorrelations among all constructs), a simple arithmetic average of the correlations is both conservative and acceptable (Tannenbaum & Cerasoli, 2013). We derived 376 effect sizes covering 55,514 respondents from 43 primary studies, representing 49 independent samples.

To explore the potential impact of research design and the quality of publication outlet on the variability of effects sizes, we tested several factors that might impact study results in a cross-classified design (HLM; Raudenbush, Bryk, Congdon, & du Toit, 2004). In the interest of space, the results are presented in an Appendix. As detailed in the Appendix, those analyses revealed that neither design features nor source quality evidenced any significant influences on the magnitude of effect sizes, indicating that methodological rigor poses no threat to the validity of our findings.

Results

Meta-analysis results are summarized for personal antecedents in Table 2, for situational antecedents in Table 3, and for outcomes in Table 4. A comprehensive breakdown of effect sizes per study appears in Table 5. Individual predispositions overall showed a positive relationship with ILBs ($\rho = .27$, $k = 12$, $N = 4432$; 80% CrI = .11, .43; 95% CI = .20–.34). In terms of research question 1, personality/propensity factors displayed a positive association with ILBs, as did general learning-related motives. As for research question 2, as shown in Table 2, several demographic factors exhibited relationships with ILBs. Specifically, age, education, sex, rank/tenure, marital status, and income all yielded confidence intervals that did not include zero.

Results are reported in Table 3 for job/task characteristics ($\rho = .21$, $k = 9$, $N = 7857$; 80% CrI = .05, .36; 95% CI = .12–.30). Results for control/autonomy, resources, and demands provided indications of a positive response to research question 3. All effects were different from zero with respect to confidence intervals, although credibility intervals spanning zero for demands suggest additional moderators. Similarly, support results are presented for support ($\rho = .32$, $k = 20$, $N = 9231$; 80% CrI = .13, .50; 95% CI = .26–.38), showing that people support, formal organizational support, and informal organizational support displayed relationships with levels of ILBs (as indicated by confidence intervals), addressing research question 4. Also in Table 3, in response to research question 5, opportunities for learning were associated with higher levels of ILBs in some cases ($\rho = .04$, $k = 8$, $N = 3342$; 80% CrI = -.21, .30; 95%

Table 2 Meta-analysis of personal antecedents of informal learning

	<i>N</i>	<i>k</i>	<i>r</i> _{obs}	<i>SD</i> _{obs}	ρ	<i>SD</i> ρ	80% CrI		% err	File drawer		Mean α		95% CI	
							.10	.90		.10	.05	<i>r</i> _{xx}	<i>r</i> _{yy}	.025	.975
Personal overall	45,947	29	.03	.09	.04	.11	-.10	.19	7.09	-20	-12	.81	.81	.00	.09
Individual predispositions	4432	12	.22	.10	.27	.12	.11	.43	22.29	14	41	.79	.80	.20	.34
Personality/propensity	3836	7	.21	.05	.27	.05	.20	.34	55.94	8	22	.76	.77	.22	.32
Learning motives	1319	8	.28	.17	.33	.20	.07	.60	15.51	14	37	.84	.83	.19	.47
Demographics	43,435	21	.01	.07	.02	.08	-.09	.12	8.95	-19	-17	.93	.80	-.04	.08
Age	38,471	15	-.06	.06	-.07	.07	-.16	.01	10.46	-24	-33	–	.76	-.11	-.03
Education	1757	5	.09	.00	.10	.00	.10	.10	100.00	-1	4	–	.72	.10	.10
Experience	1942	4	.01	.05	.01	.05	-.05	.07	49.48	-4	-3	.93	.90	-.04	.06
Female (0) vs. male (1)	38,089	13	-.02	.02	-.03	.02	-.06	.01	41.85	-16	-18	–	.76	-.05	-.01
Income	35,299	8	-.05	.03	-.05	.03	-.09	-.02	19.76	-12	-16	–	–	-.07	-.03
Rank/tenure	4363	9	.15	.13	.18	.15	-.01	.36	10.97	5	18	–	.74	.08	.28
Single (0) vs. married (1)	36,005	6	.16	.07	.17	.08	.07	.28	2.85	4	13	–	.82	.11	.23

N number of respondents, *k* number of independent samples, *r*_{obs} observed correlation after removing sampling error, *SD*_{obs} standard deviation after removing sampling error, ρ corrected population correlation, *SD* ρ corrected population standard deviation, 80% CrI the lower, upper, and range of the 80% credibility interval of the true population correlation, % error percentage of variance in the corrected population correlation accounted for by statistical artifacts (error), Filedrawer number of unpublished/unavailable studies at $\rho = .10$ or .05 needed to pull the corrected population correlation below that value, Mean α mean Cronbach's alpha reliability estimate, *r*_{xx} mean reliability of the independent variable, *r*_{yy} mean reliability of the informal learning variable, 95% CI the lower (.025) and upper (.975) bounds of the 95% confidence interval

Table 3 Meta-analysis of situational antecedents of informal learning

	<i>N</i>	<i>k</i>	<i>r</i> _{obs}	<i>SD</i> _{obs}	ρ	<i>SD</i> _{ρ}	80% CrI		% err	File Drawer		Mean α		95% CI	
							.10	.90		.10	.05	<i>r</i> _{xx}	<i>r</i> _{yy}	.025	.975
Situational overall	14,431	28	.18	.16	.22	.19	-.03	.46	7.46	22	73	.82	.81	.15	.29
Job/task characteristics	7857	9	.16	.10	.21	.12	.05	.36	13.00	5	20	.78	.78	.12	.30
Demands	5208	6	.10	.07	.13	.10	.01	.25	18.35	0	6	.73	.78	.06	.20
Resources	2040	3	.23	.10	.30	.12	.14	.46	19.02	4	11	.81	.74	.15	.45
Control/autonomy	3675	4	.25	.09	.31	.10	.17	.44	13.61	6	16	.79	.84	.20	.42
Support	9231	20	.26	.12	.32	.14	.13	.50	13.51	32	84	.83	.81	.26	.38
Formal	1734	5	.32	.23	.38	.27	.03	.73	4.83	11	27	.81	.85	.14	.62
Informal	6306	13	.24	.12	.30	.15	.11	.49	13.53	18	49	.82	.79	.22	.38
People	4595	8	.26	.09	.31	.10	.18	.45	19.52	13	34	.84	.81	.24	.38
Learning opportunities	3342	8	.03	.16	.04	.19	-.21	.30	8.83	-6	-3	.79	.82	-.11	.19
Potential for new learning	1585	2	.08	.00	.12	.00	.12	.12	100.00	0	1	.65	.75	.12	.12
Lower workload	2693	5	-.01	.17	-.02	.25	-.33	.30	6.07	-6	-6	.64	.73	-.32	.28
Time available	354	2	.12	.05	.14	.05	.07	.21	71.45	0	3	.87	.88	.06	.22

N number of respondents, *k* number of independent samples, *r*_{obs} observed correlation after removing sampling error, *SD*_{obs} standard deviation after removing sampling error, ρ corrected population correlation, *SD* _{ρ} corrected population standard deviation, 80% CrI the lower, upper, and range of the 80% credibility interval of the true population correlation, % error percentage of variance in the corrected population correlation accounted for by statistical artifacts (error), Filedrawer number of unpublished/unavailable studies at $\rho = .10$ or .05 needed to pull the corrected population correlation below that value, Mean α mean Cronbach's alpha reliability estimate, *r*_{xx} mean reliability of the independent variable, *r*_{yy} mean reliability of the informal learning variable, 95% CI the lower (.025) and upper (.975) bounds of the 95% confidence interval

CI = -.1–.19). The effect for lower workload did not include zero in the confidence interval, although potential for new learning and time available did include zero.

Findings reported in Table 4 show a generally positive link of ILBs with outcomes ($\rho = .33$, $k = 23$, $N = 10,975$; 80% CrI = .14, .51; 95% CI = .27–.39). Results seem to answer research question 6 in a positive way, such that engagement in ILBs was positively associated with positive general work attitudes. Results were robust regarding research question 7, in that engagement in informal learning was positively associated with knowledge/skill acquisition. Finally, the findings answered research question 8 in a positive way, engagement in ILBs being positively associated with performance.

Discussion

Surveys, testimonials, and expert consensus indicate that roughly 80% of workplace learning takes place informally. Yet, a mere "...20 percent of what organizations invest in learning is dedicated to enhancing informal learning" (Berg & Chyung, 2008, p. 230). We suggest this misalignment of resources exists because the current body of research has not been well-poised to provide unified, evidence-based recommendations for enhancing ILBs. The current review and meta-analyses provide an initial step towards such unification by aggregating available empirical research on informal learning

in working adults. Building on the framework outlined by Tannenbaum et al. (2010), we have detailed personal and situational antecedents of informal learning at three different levels of construct specificity. We have also articulated a rationale for exploring three outcomes of ILBs. Using this framework, we reviewed extant literature and conducted meta-analyses to examine our research questions.

With respect to antecedents, average effect sizes were comparatively the largest for situational antecedents (although, many credibility intervals still overlapped with other effect sizes). The perceived presence of support bore a positive link with the extent to which individuals reported engaging in ILBs, consistent with a substantial body of research in social psychology demonstrating the powerful impact of social cues and environments on behavior (Ross & Nisbett, 1991). Interestingly, formal sources of support (e.g., tangible rewards) averaged a stronger relationship with ILBs than informal sources of support (although credibility intervals for the relevant effect sizes overlapped). Results indicated that job/task characteristics, such as one's autonomy and available resources, are positively associated with ILBs. In contrast, results for time availability and job demands appeared more equivocal. Perhaps there exists a curvilinear relationship: too much time or low demands may obviate the need for ILBs, whereas high time pressures and demands preclude the possibility to engage in ILBs. However, that could not be tested directly in the current meta-analyses and warrants examination in future primary studies.

Table 4 Meta-analysis of outcomes of informal learning

	<i>N</i>	<i>k</i>	<i>r_{obs}</i>	<i>SD_{obs}</i>	ρ	<i>SDρ</i>	80% CrI		% err	File Drawer		Mean α		95% CI	
							.10	.90		.10	.05	<i>r_{xx}</i>	<i>r_{yy}</i>	.025	.975
Outcomes overall	10,975	23	.27	.12	.33	.15	.14	.51	12.84	39	101	.82	.82	.27	.39
Attitudes	7025	11	.23	.13	.29	.15	.09	.48	9.21	14	40	.82	.81	.19	.39
Knowledge/skill acquisition	3043	8	.33	.10	.41	.11	.26	.55	25.66	18	45	.82	.82	.32	.50
Performance	3840	10	.33	.11	.42	.13	.24	.59	19.70	23	56	.78	.79	.33	.51

N number of respondents, *k* number of independent samples, *r_{obs}* observed correlation after removing sampling error, *SD_{obs}* standard deviation after removing sampling error, ρ corrected population correlation, *SD ρ* corrected population standard deviation, 80% CrI the lower, upper, and range of the 80% credibility interval of the true population correlation, % error percentage of variance in the corrected population correlation accounted for by statistical artifacts (error), File drawer number of unpublished/unavailable studies at $\rho = .10$ or $.05$ needed to pull the corrected population correlation below that value, Mean α mean Cronbach's alpha reliability estimate, *r_{xx}* mean reliability of the independent variable, *r_{yy}* mean reliability of the informal learning variable, 95% CI the lower (.025) and upper (.975) bounds of the 95% confidence interval

Regarding personal antecedents, results were as expected for individual predispositions, which bundles together personality/propensity factors (e.g., constructs like agreeableness and extraversion), and specific learning-related motives (e.g., perceived utility of the learning activity). Common demographics including education, experience, rank, and marital status were associated with ILBs. Perhaps, as others have suggested, more educated employees may be more inclined to seek out opportunities to learn and view learning more favorably (Birdi et al., 1997; McCauley & Hezlett, 2001). The experience and rank/tenure effect sizes may reflect the fact that management employees tend to have a more favorable view of learning programs than do subordinates (McCauley & Hezlett, 2001) and may have more control over their work environment, allowing them to more readily create learning opportunities. The positive marital status results parallel meta-analyses showing marriage correlates positively with career success (Ng et al., 2005). Age and sex showed non-zero effects; however, the effects were quite small and consistent with the body of research showing little association of these characteristics with attitudes, motivation, and performance (Cerasoli, Nicklin & Ford, 2014; Ng & Feldman, 2010, 2012).

Results were strong for outcomes overall. The effect size for performance was relatively strong: engagement in ILBs explained a little under 18% of the variability in performance criteria such as rated job performance, effectiveness, salary, promotions, and project performance. Interestingly, our effect size of $\rho = .42$ outstrips the effect size found for formal training of $\rho = .30$ by Arthur et al. (2003). Put in a different metric, this would mean that those who engage in ILBs average 32% higher performance than those who do not. As a point of comparison, results of the Arthur et al. meta-analysis (2003) suggested those who undergo formal training interventions (versus those who do not) can expect a more modest 23% boost.³

It is important to note that we are not suggesting that antecedents or outcomes of ILBs will always have different effects

than those associated with FLBs; the two are likely to share many common roots and have similar outcomes. However, science and practice are still advanced by empirically testing the antecedents and outcomes of ILBs: there is value to be added by having another “tool” with which to drive organizational learning.⁴ Furthermore, the value of ILBs may be more specific to the peculiar nature of a worker's job, given the manner in which K/S are learned in context. Again, we do not suggest that FLBs and ILBs are incompatible; in fact, it would interesting to examine whether workers who do best in formal training also excel in informal training. Indeed, an investigation of whether formal training can be used to instigate ILBs would be a valuable contribution to both theory and practice. In any case, it is valuable to develop an understanding of relationships associated with the effectiveness of ILBs because they might be used in parallel, in conjunction with, or even independently from FLBs.

Finally, as noted in the introduction and throughout the review, much of this work is intended to take a preliminary, integrating approach to ILBs and their role in organizations. We developed a broad definition of ILBs and a general framework with which to address antecedents and outcomes. Through our review, we found ILBs discussed in many disciplines, and research on informal learning spans a broad research base. One reason for taking the broad approach we did is to ensure we could capture the important components from multiple disciplines, reconciling multiple definitions (e.g., the Tannenbaum et al. versus the Noe et al. frameworks). Another reason for the broad approach is that there are many different constructs that could fall under the umbrella of ILBs (e.g., action learning), some of which have already been meta-analyzed (e.g., debriefs;

³ Arthur et al. report a sample-weighted mean *d* of .62 for both behavioral and result criteria. To interpret their findings against ours, we converted to a correlation, yielding $\rho = .30$. Then, using a cumulative normal distribution table, $\rho = .42 \rightarrow d = .93 \rightarrow 32\%$; $\rho = .30 \rightarrow d = .62 \rightarrow 23\%$

⁴ We thank an anonymous reviewer for this point.

Table 5 Breakdown of studies in the meta-analysis

	Author	Year	<i>r</i>	<i>N</i>	α IV	α DV	IV description	IV1	IV2	IV3
1	Bednall et al.	2014	-.17	238		.80	Sex	P	DM	SEX
2	Bednall et al.	2014	-.19	238		.80	Tenure	P	DM	RNK
3	Bednall et al.	2014	.09	238		.85	Hours worked	S	OP	WRK
4	Bednall et al.	2014	-.11	238		.81	Sex	P	DM	SEX
5	Bednall et al.	2014	-.12	238		.81	Tenure	P	DM	RNK
6	Bednall et al.	2014	.14	238		.81	Hours worked	S	OP	WRK
7	Berg and Chyung	2008	.05	125			Learning culture (overall)	S	SU	INF
8	Berg and Chyung	2008	.02	125			Learning culture (individual)	S	SU	INF
9	Berg and Chyung	2008	.10	125			Learning culture (team)	S	SU	INF
10	Berg and Chyung	2008	.06	125			Learning culture (organizational)	S	SU	INF
11	Berg and Chyung	2008	.20	125			Age	P	DM	AGE
12	Berg and Chyung	2008	.11	125			Age	P	DM	AGE
13	Berg and Chyung	2008	-.06	125			Age	P	DM	AGE
14	Berg and Chyung	2008	.05	125			Age	P	DM	AGE
15	Berg and Chyung	2008	-.08	125			Age	P	DM	AGE
16	Berg and Chyung	2008	.34	125			Age	P	DM	AGE
17	Berg and Chyung	2008	.25	125			Age	P	DM	AGE
18	Berg and Chyung	2008	.06	125			Age	P	DM	AGE
19	Berg and Chyung	2008	.10	125			Age	P	DM	AGE
20	Berg and Chyung	2008	-.04	110			Sex (female vs. male)	P	DM	SEX
21	Berg and Chyung	2008	-.03	110			Education	P	DM	EDU
22	Bickmore	2011	.26	127			Effectiveness (responsiveness to faculty)	O	PF	
23	Bickmore	2011	.40	128			Effectiveness (responsiveness to students)	O	PF	
24	Bickmore	2011	.30	122			Effectiveness (responsiveness to schools)	O	PF	
25	Carson	2013	.31	780		.86	GPA (performance)	O	PF	
26	Carson	2013	.13	703		.86	GPA (final grades)	O	PF	
27	Choi and Jacobs	2011	.28	203	.81	.69	Self-efficacy	P	IP	PPP
28	Choi and Jacobs	2011	.49	203	.88	.69	Learning goal orientation	P	IP	LRN
29	Choi and Jacobs	2011	.34	203	.88	.69	Motivation to learn	P	IP	LRN
30	Choi and Jacobs	2011	.24	203	.77	.69	Organizational support	S	SU	INF
31	Choi and Jacobs	2011	.28	203	.73	.69	Supervisor support	S	SU	PPL
32	Choi and Jacobs	2011	.30	203	.64	.69	Job characteristics (demands)	S	JT	DEM
33	Choi and Jacobs	2011	.28	203	.81	.76	Self-efficacy	P	IP	PPP
34	Choi and Jacobs	2011	.44	203	.88	.76	Learning goal orientation	P	IP	LRN
35	Choi and Jacobs	2011	.40	203	.88	.76	Motivation to learn	P	IP	LRN

Table 5 (continued)

	Author	Year	<i>r</i>	<i>N</i>	α IV	α DV	IV description	IV1	IV2	IV3
36	Choi and Jacobs	2011	.15	203	.77	.76	Organizational support	S	SU	INF
37	Choi and Jacobs	2011	.11	203	.73	.76	Supervisor support	S	SU	PPL
38	Choi and Jacobs	2011	.14	203	.64	.76	Job characteristics (demands)	S	JT	DEM
39	Choi and Jacobs	2011	.23	203	.81	.68	Self-efficacy	P	IP	PPP
40	Choi and Jacobs	2011	.31	203	.88	.68	Learning goal orientation	P	IP	LRN
41	Choi and Jacobs	2011	.23	203	.88	.68	Motivation to learn	P	IP	LRN
42	Choi and Jacobs	2011	.12	203	.77	.68	Organizational support	S	SU	INF
43	Choi and Jacobs	2011	.06	203	.73	.68	Supervisor support	S	SU	PPL
44	Choi and Jacobs	2011	.21	203	.64	.68	Job characteristics (demands)	S	JT	DEM
45	De Groot et al.	2012	.31	1290	.60	.62	Need to be informed	P	IP	PPP
46	De Groot et al.	2012	.23	1290	.60	.61	Need to be informed	P	IP	PPP
47	De Groot et al.	2012	.27	1290	.60	.64	Need to be informed	P	IP	PPP
48	De Groot et al.	2012	.20	1290	.60	.60	Need to be informed	P	IP	PPP
49	De Groot et al.	2012	.19	1290	.62	.62	Knowledge base stability	S	JT	DEM
50	De Groot et al.	2012	.12	1290	.62	.61	Knowledge base stability	S	JT	RES
51	De Groot et al.	2012	.15	1290	.62	.64	Knowledge base stability	S	JT	RES
52	De Groot et al.	2012	.18	1290	.62	.60	Knowledge base stability	S	JT	RES
53	De Groot et al.	2012	-.02	1290	.64	.62	Heavy workload/lack time	S	OP	WRK
54	De Groot et al.	2012	-.02	1290	.64	.61	Heavy workload/lack time	S	OP	WRK
55	De Groot et al.	2012	-.03	1290	.64	.64	Heavy workload/lack time	S	OP	WRK
56	De Groot et al.	2012	-.03	1290	.64	.60	Heavy workload/lack time	S	OP	WRK
57	De Groot et al.	2012	.07	1290	.65	.62	Opportunities for feedback	S	OP	POT
58	De Groot et al.	2012	.16	1290	.65	.61	Opportunities for feedback	S	OP	POT
59	De Groot et al.	2012	.03	1290	.65	.64	Opportunities for feedback	S	OP	POT
60	De Groot et al.	2012	.01	1290	.65	.60	Opportunities for feedback	S	OP	POT
61	Digby	2010	.04	1000			Sex (female vs. male)	P	DM	SEX
62	Digby	2010	.11	1000			Education	P	DM	EDU
63	Digby	2010	-.10	1000			Age	P	DM	AGE
64	Digby	2010	.13	1000			Income	P	DM	INC
65	Digby	2010	.24	1000	.73		Knowledge	O	KS	
66	Digby	2010	.19	1000	.79		Attitudes	O	AT	
67	Digby	2010	.40	1000	.62		Performance	O	PF	
68	Enos et al.	2003	.06	84	.89	.93	Proficiency	O	KS	
69	Enos et al.	2003	-.15	84	.93	.93	Coworker support	S	SU	PPL
70	Enos et al.	2003	-.26	84	.95	.93	Supervisor transfer Support	S	SU	PPL

Table 5 (continued)

	Author	Year	<i>r</i>	<i>N</i>	α IV	α DV	IV description	IV1	IV2	IV3
71	Enos et al.	2003	-.46	84	.85	.93	Organizational support for transfer	S	SU	FOR
72	Enos et al.	2003	-.01	84	.87	.93	Transfer of learning	O	KS	
73	Froelich et al.	2014	.32	143	.80	.61	Organizational learning culture	S	SU	INF
74	Froelich et al.	2014	.10	143	.93	.61	Job appraisal performance	O	PF	
75	Froelich et al.	2014	.27	143	.62	.61	Core skills	O	KS	
76	Froelich et al.	2014	.38	143	.70	.61	Perceived career development	O	PF	
77	Froelich et al.	2014	.28	143	.76	.61	Subjective job performance	O	PF	
78	Gijbels et al.	2010	.29	115	.72	.83	Job demands	S	JT	DEM
79	Gijbels et al.	2010	.12	115	.79	.83	Job control	S	JT	C/A
80	Gijbels et al.	2010	.12	115	.79	.83	Social support	S	SU	
81	Gijbels et al.	2010	.49	115	.83	.83	Self-learning orientation	P	IP	LRN
82	Gijbels et al.	2012	.33	73	.81	.92	Job demands	S	JT	DEM
83	Gijbels et al.	2012	.36	73	.87	.92	Job control	S	JT	C/A
84	Gijbels et al.	2012	.05	73	.79	.92	Social support	S	SU	PPL
85	Gijbels et al.	2012	.61	73	.83	.92	Self-learning orientation	P	IP	LRN
86	Gonzales	1985	.24	170			Degree work involvement	O	PF	
87	Gonzales	1985	.18	170			Experience	P	DM	RNK
88	Gonzales	1985	.04	170			Experience	P	DM	EXP
89	Gonzales	1985	.13	170			Experience	P	DM	EXP
90	Gonzales	1985	.26	170			Age	P	DM	AGE
91	Gonzales	1985	-.04	170			Income	P	DM	INC
92	Gonzales	1985	.24	170			Degree Work Involvement	O	PF	
93	Gonzales	1985	.22	170			Experience	P	DM	RNK
94	Gonzales	1985	-.01	170			Experience	P	DM	EXP
95	Gonzales	1985	.17	170			Experience	P	DM	EXP
96	Gonzales	1985	.30	170			Age	P	DM	AGE
97	Gonzales	1985	-.09	170			Income	P	DM	INC
98	Gonzales	1985	.18	170			Degree Work Involvement	O	PF	
99	Gonzales	1985	.16	170			Experience	P	DM	RNK
100	Gonzales	1985	-.02	170			Experience	P	DM	EXP
101	Gonzales	1985	.17	170			Experience	P	DM	EXP
102	Gonzales	1985	.40	170			Age	P	DM	AGE
103	Gonzales	1985	-.27	170			Income	P	DM	INC
104	Gonzales	1985	.17	170			Degree work involvement	O	PF	
105	Gonzales	1985	.87	170			Experience	P	DM	RNK

Table 5 (continued)

	Author	Year	<i>r</i>	<i>N</i>	α IV	α DV	IV description	IV1	IV2	IV3
106	Gonzales	1985	.10	170			Experience	P	DM	EXP
107	Gonzales	1985	.10	170			Experience	P	DM	EXP
108	Gonzales	1985	.28	170			Age	P	DM	AGE
109	Gonzales	1985	-.06	170			Income	P	DM	INC
110	Gonzales	1985	.05	170			Sex (female vs. male)	P	DM	SEX
111	Gonzales	1985	-.04	170			Sex (female vs. male)	P	DM	SEX
112	Gonzales	1985	-.11	170			Sex (female vs. male)	P	DM	SEX
113	Gonzales	1985	.08	170			Sex (female vs. male)	P	DM	SEX
114	Hicks et al.	2007	-.11	110			Rank (trainee vs. manager)	P	DM	RNK
115	Hicks et al.	2007	-.14	100			Rank (trainee vs. partner)	P	DM	RNK
116	Hicks et al.	2007	-.04	74			Rank (manager v. partner)	P	DM	RNK
117	Hicks et al.	2007	.01	109			Rank (trainee vs. manager)	P	DM	RNK
118	Hicks et al.	2007	.15	100			Rank (trainee vs. partner)	P	DM	RNK
119	Hicks et al.	2007	.15	73			Rank (manager v. partner)	P	DM	RNK
120	Hicks et al.	2007	.21	109			Rank (trainee vs. manager)	P	DM	RNK
121	Hicks et al.	2007	.32	100			Rank (trainee vs. partner)	P	DM	RNK
122	Hicks et al.	2007	.16	73			Rank (manager vs. partner)	P	DM	RNK
123	Hicks et al.	2007	-.25	106			Rank (trainee vs. manager)	P	DM	RNK
124	Hicks et al.	2007	-.41	98			Rank (trainee vs. partner)	P	DM	RNK
125	Hicks et al.	2007	-.13	72			Rank (manager vs partner)	P	DM	RNK
126	Hicks et al.	2007	.00	109			Rank (trainee vs. manager)	P	DM	RNK
127	Hicks et al.	2007	-.08	100			Rank (trainee vs. partner)	P	DM	RNK
128	Hicks et al.	2007	-.09	73			Rank (manager vs. partner)	P	DM	RNK
129	Houde	2014	.49	26			Support (accessible)	S	SU	
130	Houde	2014	.34	26			Support (knowledgeable)	S	SU	
131	Houde	2014	.33	26			Support (trusted source of IL)	S	SU	
132	Houde	2014	.10	26			Sex	P	DM	SEX
133	Houde	2014	-.03	26			Age	P	DM	AGE
134	Hurns	2013	.20	86	.71	.81	Critically reflective nature	P	IP	LRN
135	Hutchens et al.	2010	-.19	107	.89	.78	Research literature	P	IP	LRN
136	Hutchens et al.	2010	.00	107		.78	Education	P	DM	EDU
137	Hutchens et al.	2010	.08	107		.78	Job level	P	DM	RNK
138	Jeon and Kim	2012	.24	1899	.88		Organizational leadership	S	SU	PPL
139	Jeon and Kim	2012	.26	1899	.85		Open communication	S	SU	INF
140	Jeon and Kim	2012	.18	1899	.69		Innovative culture	S	SU	INF

Table 5 (continued)

	Author	Year	<i>r</i>	<i>N</i>	α IV	α DV	IV description	IV1	IV2	IV3
141	Jeon and Kim	2012	.15	1899			Non-routineness of task	S	JT	C/A
142	Jeon and Kim	2012	.18	1899			Task satisfaction	O	AT	
143	Jeon and Kim	2012	.17	1899	.88		Organizational leadership	S	SU	PPL
144	Jeon and Kim	2012	.20	1899	.85		Open communication	S	SU	INF
145	Jeon and Kim	2012	.12	1899	.69		Innovative culture	S	SU	INF
146	Jeon and Kim	2012	.18	1899			Non-routineness of task	S	JT	C/A
147	Jeon and Kim	2012	.16	1899			Task satisfaction	O	AT	
148	Lindner (1995 sample)	2012	-.04	12,109			Age	P	DM	AGE
149	Lindner (1995 sample)	2012	.00	12,109			Sex (female vs. male)	P	DM	SEX
150	Lindner (1995 sample)	2012	.09	12,109			Marital status	P	DM	MAR
151	Lindner (1995 sample)	2012	-.06	12,109			Earnings	P	DM	INC
152	Lindner (1999 sample)	2012	-.04	4108			Age	P	DM	AGE
153	Lindner (1999 sample)	2012	-.03	4108			Sex (female vs. male)	P	DM	SEX
154	Lindner (1999 sample)	2012	.12	4108			Marital status	P	DM	MAR
155	Lindner (1999 sample)	2012	-.06	4108			Earnings	P	DM	INC
156	Lindner (2001 sample)	2012	-.09	5886			Age	P	DM	AGE
157	Lindner (2001 sample)	2012	-.05	5886			Sex (female vs. male)	P	DM	SEX
158	Lindner (2001 sample)	2012	.24	5886			Marital status	P	DM	MAR
159	Lindner (2001 sample)	2012	-.04	5886			Earnings	P	DM	INC
160	Lindner (2003 sample)	2012	-.09	7396			Age	P	DM	AGE
161	Lindner (2003 sample)	2012	-.02	7396			Sex (female vs. male)	P	DM	SEX
162	Lindner (2003 sample)	2012	.21	7396			Marital status	P	DM	MAR
163	Lindner (2003 sample)	2012	-.06	7396			Earnings	P	DM	INC
164	Lindner (2005 sample)	2012	-.09	4567			Age	P	DM	AGE
165	Lindner (2005 sample)	2012	-.03	4567			Sex (female vs. male)	P	DM	SEX
166	Lindner (2005 sample)	2012	.24	4567			Marital status	P	DM	MAR
167	Lindner (2005 sample)	2012	-.08	4567			Earnings	P	DM	INC
168	Livingstone	2001	-.21	489			Hours worked by men	S	OP	WRK
169	Livingstone	2001	-.10	390			Hours worked by women	S	OP	WRK
170	Livingstone and Raykov	2008	.22	2895			Decision maker?	P	DM	RNK
171	Livingstone and Stowe	2007	-.31	286			Age	P	DM	AGE
172	Livingstone and Stowe	2007	-.77	286			Age	P	DM	AGE
173	Livingstone and Stowe	2007	.91	286			Paid hours worked	S	OP	WRK
174	Livingstone and Stowe	2007	-.06	286			Paid hours worked	S	OP	WRK
175	Maringka	2014	-.17	477		.66	Sex	P	DM	SEX

Table 5 (continued)

	Author	Year	<i>r</i>	<i>N</i>	α IV	α DV	IV description	IV1	IV2	IV3
176	Maringka	2014	.09	477		.66	Tenure: time in organization	P	DM	RNK
177	Maringka	2014	.37	477	.87	.66	Support from management	S	SU	PPL
178	Maringka	2014	.40	477	.82	.66	Support from peers	S	SU	PPL
179	Maringka	2014	.38	477	.77	.66	Supportive organizational culture	S	SU	INF
180	Maringka	2014	.37	477	.90	.66	Access to work resources	S	JT	RES
181	Maringka	2014	.39	477	.89	.66	Engagement	O	AT	
182	Maringka	2014	.25	477	.78	.66	Performance	O	PF	
183	Maringka	2014	.16	477		.66	Age	P	DM	AGE
184	Maringka	2014	.10	477		.66	Education level	P	DM	EDU
185	Maringka	2014	.10	477		.66	Tenure: level in organization	P	DM	RNK
186	Matsuo and Nakahara	2013	.62	127	.96	.91	Supervisor learning support	S	SU	INF
187	Matsuo and Nakahara	2013	.34	127	.85	.91	Structure for problem solving present	S	SU	FOR
188	Matsuo and Nakahara	2013	.45	127	.86	.91	Supervisor empowerment	S	SU	INF
189	Matsuo and Nakahara	2013	.66	127	.91	.91	Knowledge acquisition, sharing	O	KS	
190	Moon and Na	2009	.40	388	.91	.85	Learning competency	O	KS	
191	Moon and Na	2009	.35	388	.84	.85	Motivation to learn	P	IP	LRN
192	Moon and Na	2009	.28	388	.78	.85	curiosity	P	IP	PPP
193	Moon and Na	2009	.16	388	.57	.85	Self-esteem	P	IP	PPP
194	Moon and Na	2009	.11	388	.66	.85	Locus of control	P	IP	PPP
195	Moon and Na	2009	.50	388		.85	Merit system	S	SU	FOR
196	Moon and Na	2009	.55	388		.85	Communication	S	SU	INF
197	Noe et al.	2013	.00	180		.71	Age	P	DM	AGE
198	Noe et al.	2013	−.09	180		.71	Tenure	P	DM	RNK
199	Noe et al.	2013	.16	180	.87	.71	Conscientiousness	P	IP	PPP
200	Noe et al.	2013	.20	180	.76	.71	Emotional stability	P	IP	PPP
201	Noe et al.	2013	.16	180	.77	.71	Openness to experience	P	IP	PPP
202	Noe et al.	2013	.17	180	.80	.71	Agreeableness	P	IP	PPP
203	Noe et al.	2013	.26	180	.71	.71	Extraversion	P	IP	PPP
204	Noe et al.	2013	.17	180	.70	.71	General self-efficacy	P	IP	PPP
205	Noe et al.	2013	.31	180	.69	.71	Zest	P	IP	PPP
206	Ouweneel et al.	2009	.08	1588	.86	.78	Job demands	S	JT	DEM
207	Ouweneel et al.	2009	.35	1588	.71	.78	Job control	S	JT	C/A
208	Ouweneel et al.	2009	.37	1588	.92	.78	Support (supervisor)	S	SU	PPL
209	Ouweneel et al.	2009	.24	1588	.76	.78	Support (colleague)	S	SU	PPL
210	Parise and Spillane	2010	.14	1418	.94		Supportive learning community	S	SU	INF

Table 5 (continued)

	Author	Year	<i>r</i>	<i>N</i>	α IV	α DV	IV description	IV1	IV2	IV3
211	Parise and Spillane	2010	.27	1418	.94		Supportive learning community	S	SU	INF
212	Parise and Spillane	2010	.17	1418	.94	.90	Supportive learning community	S	SU	INF
213	Parise and Spillane	2010	.15	1418	.94		Supportive learning community	S	SU	INF
214	Parise and Spillane	2010	-.04	1418			Years of experience	P	DM	EXP
215	Parise and Spillane	2010	.05	1418			Years of experience	P	DM	EXP
216	Parise and Spillane	2010	-.05	1418		.90	Years of experience	P	DM	EXP
217	Parise and Spillane	2010	-.08	1418			Years of experience	P	DM	EXP
218	Parise and Spillane	2010	.16	1418	.70		Teacher self-efficacy	P	IP	PPP
219	Parise and Spillane	2010	.21	1418	.70	.90	Teacher self-efficacy	P	IP	PPP
220	Parise and Spillane	2010	.12	1418	.70		Teacher self-efficacy	P	IP	PPP
221	Parise and Spillane	2010	.09	1418	.70		Teacher self-efficacy	P	IP	PPP
222	Pike	1999	.09	295		.84	Residence: interactive?	S	OP	POT
223	Pike	1999	.16	295		.89	Residence: interactive?	S	OP	POT
224	Reardon	2010	.28	288			Learning culture	S	SU	INF
225	Reychav and Te'eni	2009	.52	273	.85	.94	Utility of knowledge sharing	O	AT	RES
226	Reychav and Te'eni	2009	.38	273	.92	.94	(utility) IT for knowledge management	S	JT	RES
227	Riaz et al.	2010	.41	473	.82	.74	Web-based learning	O	KS	AGE
228	Richter et al.	2011	-.09	1939		.82	Age	P	DM	AGE
229	Richter et al.	2011	.06	1939			Age	P	DM	AGE
230	Richter et al.	2011	-.18	1939		.82	Sex (female vs. male)	P	DM	SEX
231	Richter et al.	2011	.07	1939			Sex (female vs. male)	P	DM	SEX
232	Richter et al.	2011	.02	1939		.82	Marital status	P	DM	MAR
233	Richter et al.	2011	-.02	1939			Marital status	P	DM	MAR
234	Richter et al.	2011	.15	1939	.75	.82	Work engagement	O	AT	
235	Richter et al.	2011	.11	1939	.75		Work engagement	O	AT	
236	Richter et al.	2011	-.04	1939		.82	Service responsibilities?	S	JT	DEM
237	Richter et al.	2011	.06	1939			Service responsibilities?	S	JT	DEM
238	Richter et al.	2011	.02	1939		.82	Management responsibilities?	S	JT	DEM
239	Richter et al.	2011	.03	1939			Management responsibilities?	S	JT	DEM
240	Rowden	2002	.29	794	.88	.73	Supportive environment	S	SU	
241	Rowden	2002	.58	794	.76	.73	Recognition	S	SU	
242	Rowden	2002	.28	794	.75	.73	Affective enjoyment	O	AT	
243	Rowden	2002	.20	794	.72	.73	Compensation	S	SU	FOR
244	Rowden	2002	.44	794	.87	.73	Overall job satisfaction	O	AT	
245	Rowden and Conine	2005	.65	341	.80	.83	Recognition	S	SU	FOR

Table 5 (continued)

	Author	Year	<i>r</i>	<i>N</i>	α IV	α DV	IV description	IV1	IV2	IV3
246	Rowden and Conine	2005	.48	341	.83	.83	Compensation	S	SU	FOR
247	Rowden and Conine	2005	.45	341	.74	.83	Climate	S	SU	INF
248	Rowden and Conine	2005	.49	341	.74	.83	Contentment with work/organization	O	AT	
249	Rowden and Conine	2005	.64	341	.86	.83	Overall job satisfaction	O	AT	
250	Sanders et al.	2011	.19	132	.94	.82	Perceived behavioral control	P	IP	PPP
251	Sanders et al.	2011	.24	132	.84	.82	Attitude/expected value	O	AT	
252	Sanders et al.	2011	.28	132	.90	.82	Subjective norms	S	SU	PPL
253	Sanders et al.	2011	-.12	132	.80	.82	Job insecurity	O	AT	
254	Sanders et al.	2011	.41	132	.84	.82	Management support	S	SU	PPL
255	Sanders et al.	2011	.15	132	.70	.82	Coworker support	S	SU	PPL
256	Sanders et al.	2011	.11	132	.89	.82	Career orientation	P	IP	LRN
257	Santos and Ali (a)	2012	.79	15			Utility of learning activity	O	AT	
258	Santos and Ali (a)	2012	.35	15			Utility of learning activity	O	AT	
259	Santos and Ali (a)	2012	.66	15			Utility of learning activity	O	AT	
260	Santos and Ali (a)	2012	.53	15			Utility of learning activity	O	AT	
261	Santos and Ali (a)	2012	.71	15			Utility of learning activity	O	AT	
262	Santos and Ali (a)	2012	.71	15			Utility of learning activity	O	AT	
263	Santos and Ali (a)	2012	.71	14			Utility of learning activity	O	AT	
264	Santos and Ali (a)	2012	.62	13			Utility of learning activity	O	AT	
265	Santos and Ali (a)	2012	.69	15			Utility of learning activity	O	AT	
266	Santos and Ali (b)	2012	.45	16			Utility of learning activity	O	AT	
267	Santos and Ali (b)	2012	.66	16			Utility of learning activity	O	AT	
268	Santos and Ali (b)	2012	.71	16			Utility of learning activity	O	AT	
269	Santos and Ali (b)	2012	.59	16			Utility of learning activity	O	AT	
270	Santos and Ali (b)	2012	.60	16			Utility of learning activity	O	AT	
271	Santos and Ali (b)	2012	.62	16			Utility of learning activity	O	AT	
272	Santos and Ali (b)	2012	.68	16			Utility of learning activity	O	AT	
273	Santos and Ali (b)	2012	.54	16			Utility of learning activity	O	AT	
274	Santos and Ali (b)	2012	.62	16			Utility of learning activity	O	AT	
275	Santos and Ali (b)	2012	.63	16			Utility of learning activity	O	AT	
276	Santos and Ali (b)	2012	.55	16			Utility of learning activity	O	AT	
277	Santos and Ali (b)	2012	.67	16			Utility of learning activity	O	AT	
278	Santos and Ali (b)	2012	.74	16			Utility of learning activity	O	AT	
279	Santos and Ali (b)	2012	.50	16			Utility of learning activity	O	AT	
280	Santos and Ali (b)	2012	.65	16			Utility of learning activity	O	AT	

Table 5 (continued)

	Author	Year	<i>r</i>	<i>N</i>	α IV	α DV	IV description	IV1	IV2	IV3
281	Santos and Ali (b)	2012	.60	16			Utility of learning activity	O	AT	
282	Santos and Ali (b)	2012	.51	16			Utility of learning activity	O	AT	
283	Santos and Ali (b)	2012	.11	16			Utility of learning activity	O	AT	
284	Santos and Ali (b)	2012	.20	16			Utility of learning activity	O	AT	
285	Scheuringer	2013	.36	225	.87	.90	Core self-evaluation	P	IP	PPP
286	Spreitzer et al. (1)	1997	.57	772		.85	Current performance	O	PF	
287	Spreitzer et al. (1)	1997	.40	772		.85	Executive potential	O	PF	
288	Spreitzer et al. (1)	1997	.58	772		.85	Job content learning	O	KS	
289	Spreitzer et al. (1)	1997	.54	772		.85	Behavioral skill learning	O	KS	
290	Spreitzer et al. (1)	1997	.53	772		.85	Success with international issues	O	PF	
291	Spreitzer et al. (1)	1997	.05	772		.85	Interpersonal skills	O	KS	
292	Spreitzer et al. (1)	1997	.20	772		.85	Technical skills	O	KS	
293	Spreitzer et al. (2)	1997	.51	56		.85	Current performance	O	PF	
294	Spreitzer et al. (2)	1997	.05	56		.85	Leadership and Motivation	O	PF	
295	Spreitzer et al. (2)	1997	.20	56	.90	.85	Strategic business awareness	O	PF	
296	Spreitzer et al. (2)	1997	.38	56	.80	.85	Achievement of results	O	PF	
297	Spreitzer et al. (2)	1997	.35	56		.85	Executive potential	O	PF	
298	Spreitzer et al. (2)	1997	.46	56		.85	Job content learning	O	KS	
299	Spreitzer et al. (2)	1997	.49	56		.85	Behavioral skill learning	O	KS	
300	Spreitzer et al. (2)	1997	.53	56		.85	Success with international issues	O	PF	
301	van der Heijden et al.	2009	.33	215	.88	.93	Time	S	OP	TIM
302	van der Heijden et al.	2009	-.13	215	.88	.89	Time	S	OP	TIM
303	van der Heijden et al.	2009	-.01	215	.88	.84	Time	S	OP	TIM
304	van der Heijden et al.	2009	.01	215	.88	.89	Time	S	OP	TIM
305	van der Heijden et al.	2009	.44	215	.84	.93	External support (team)	S	SU	INF
306	van der Heijden et al.	2009	-.04	215	.84	.89	External support (team)	S	SU	INF
307	van der Heijden et al.	2009	.12	215	.84	.84	External support (team)	S	SU	INF
308	van der Heijden et al.	2009	.04	215	.84	.89	External support (team)	S	SU	INF
309	van der Heijden et al.	2009	.19	215	.93	.93	Expertise	P	DM	EXP
310	van der Heijden et al.	2009	-.02	215	.93	.89	Expertise	P	DM	EXP
311	van der Heijden et al.	2009	.20	215	.93	.84	Expertise	P	DM	EXP
312	van der Heijden et al.	2009	.17	215	.93	.89	Expertise	P	DM	EXP
313	van der Heijden et al.	2009	.15	215	.83	.93	Anticipation and optimization	P	IP	LRN
314	van der Heijden et al.	2009	.23	215	.83	.89	Anticipation and optimization	P	IP	LRN
315	van der Heijden et al.	2009	.15	215	.83	.84	Anticipation and optimization	P	IP	LRN

Table 5 (continued)

	Author	Year	<i>r</i>	<i>N</i>	α IV	α DV	IV description	IV1	IV2	IV3
316	van der Heijden et al.	2009	.29	215	.83	.89	Anticipation and optimization	P	IP	LRN
317	van der Heijden et al.	2009	.13	215	.81	.93	Personal flexibility	O	PF	
318	van der Heijden et al.	2009	.03	215	.81	.89	Personal flexibility	O	PF	
319	van der Heijden et al.	2009	.20	215	.81	.84	Personal flexibility	O	PF	
320	van der Heijden et al.	2009	.19	215	.81	.89	Personal flexibility	O	PF	
321	van der Klink et al.	2014	-.08	139		.92	Age	P	DM	AGE
322	van der Klink et al.	2014	-.16	139		.86	Age	P	DM	AGE
323	van der Klink et al.	2014	-.24	139		.84	Age	P	DM	AGE
324	van der Klink et al.	2014	-.26	139		.84	Age	P	DM	AGE
325	van der Klink et al.	2014	-.15	139		.92	Tenure	P	DM	RNK
326	van der Klink et al.	2014	-.18	139		.86	Tenure	P	DM	RNK
327	van der Klink et al.	2014	-.12	139		.84	Tenure	P	DM	RNK
328	van der Klink et al.	2014	-.26	139		.84	Tenure	P	DM	RNK
329	van der Klink et al.	2014	.17	139		.92	Salary	O	PF	
330	van der Klink et al.	2014	.29	139		.86	Salary	O	PF	
331	van der Klink et al.	2014	.09	139		.84	Salary	O	PF	
332	van der Klink et al.	2014	.13	139		.84	Salary	O	PF	
333	van der Klink et al.	2014	.03	139		.92	Promotions	O	PF	
334	van der Klink et al.	2014	.11	139		.86	Promotions	O	PF	
335	van der Klink et al.	2014	.14	139		.84	Promotions	O	PF	
336	van der Klink et al.	2014	.05	139		.84	Promotions	O	PF	
337	van der Klink et al.	2014	.36	139	.85	.92	Time	S	OP	TIM
338	van der Klink et al.	2014	.30	139	.85	.86	Time	S	OP	TIM
339	van der Klink et al.	2014	.11	139	.85	.84	Time	S	OP	TIM
340	van der Klink et al.	2014	.14	139	.85	.84	Time	S	OP	TIM
341	van der Klink et al.	2014	.61	139	.86	.92	Peer support (learning climate)	S	SU	PPL
342	van der Klink et al.	2014	.36	139	.86	.86	Peer support (learning climate)	S	SU	PPL
343	van der Klink et al.	2014	.41	139	.86	.84	Peer support (learning climate)	S	SU	PPL
344	van der Klink et al.	2014	.18	139	.86	.84	Peer support (learning climate)	S	SU	PPL
345	van der Klink et al.	2014	.05	139	.92	.92	Expertise	P	DM	EXP
346	van der Klink et al.	2014	.11	139	.92	.86	Expertise	P	DM	EXP
347	van der Klink et al.	2014	.09	139	.92	.84	Expertise	P	DM	EXP
348	van der Klink et al.	2014	.19	139	.92	.84	Expertise	P	DM	EXP
349	van der Klink et al.	2014	.12	139	.82	.92	Anticipation and optimization	P	AT	
350	van der Klink et al.	2014	.33	139	.82	.86	Anticipation and optimization	P	AT	

Table 5 (continued)

	Author	Year	<i>r</i>	<i>N</i>	αIV	αDV	IV description	IV1	IV2	IV3
351	van der Klink et al.	2014	.20	139	.82	.84	Anticipation and optimization	P	AT	
352	van der Klink et al.	2014	.35	139	.82	.84	Anticipation and optimization	P	AT	
353	van der Klink et al.	2014	.13	139	.79	.92	Flexibility	O	PF	
354	van der Klink et al.	2014	.18	139	.79	.86	Flexibility	O	PF	
355	van der Klink et al.	2014	.23	139	.79	.84	Flexibility	O	PF	
356	van der Klink et al.	2014	.36	139	.79	.84	Flexibility	O	PF	
357	van der Klink et al.	2014	.26	139	.81	.92	Engagement (corporate sense)	O	AT	
358	van der Klink et al.	2014	.39	139	.81	.86	Engagement (corporate sense)	O	AT	
359	van der Klink et al.	2014	.50	139	.81	.84	Engagement (corporate sense)	O	AT	
360	van der Klink et al.	2014	.32	139	.81	.84	Engagement (corporate sense)	O	AT	
361	Wasiyo	2010	.11	656	.83	.71	Social capital	S	SU	INF
362	Wasiyo	2010	.12	656	.88	.71	Project performance	O	PF	
363	Wasiyo	2010	.34	656	.76	.71	Project manager effectiveness	O	PF	
364	Wasiyo	2010	.31	656	.75	.71	Improved project effectiveness	O	PF	
365	Welch	2013	-.05	63			Tenure	P	DM	RNK
366	Welch	2013	-.22	63			Tenure	P	DM	RNK
367	Welch	2013	.01	63			Experience	P	DM	EXP
368	Welch	2013	.09	63			Experience	P	DM	EXP
369	Welch	2013	.17	63			Sex	P	DM	SEX
370	Welch	2013	-.05	63			Sex	P	DM	SEX
371	Welch	2013	.03	63			Age	P	DM	AGE
372	Welch	2013	-.03	63			Age	P	DM	AGE
373	Welch	2013	.11	63			Income	P	DM	INC
374	Welch	2013	-.02	63			Income	P	DM	INC
375	Welch	2013	.05	63			Education	P	DM	EDU
376	Welch	2013	-.13	63			Education	P	DM	EDU

r observed/calculated correlation between informal learning and an antecedent; αIV and αDV observed reliabilities of the focal variable and informal learning, respectively; *IV1* level 1 category (*P* person, *S* situation, *O* outcomes); *IV2* level 2 category (*IP* individual predispositions, *DM* demographics, *JT* job/task characteristics, *SU* support, *OP* opportunity for learning, *ATT* attitudes, *K&S* knowledge and skill, *PER* = performance); *IV3* level 3 category (*PPP* positive personality/propensities, *PLM* positive learning-related motives, *AGE* age, *EDU* education, *SEX* sex, *INC* income, *RNK* rank/tenure, *EXP* experience, *MAR* marital status, *DEM* demands, *RES* resources, *TIM* time, *C/A* control/autonomy, *INF* informal support, *FOR* formal support, *PEO* support from people, *POT* potential for new learning opportunities, *WRK* lower workload, *TIM* time available), ? a dichotomous or yes/no variable

Tannenbaum & Cerasoli, 2013), and it would be impossible to perform more than a cursory exploration of each in a combined meta-analysis. The final, and perhaps most important, reason for our broad approach is driven by the purpose of our research. That is, it is our belief that a broader review will be more effective at generating interest, research, and pragmatic advice surrounding informal learning in working adults than would be a comprehensive review that explores every individual construct that could fall under the umbrella of ILBs. A comprehensive, deep-level review might reduce the incentive to pursue future research on ILBs in primary data (Chan & Arvey, 2012) and in our opinion would not be possible at this time, given the limited nature of extant research on this topic. A deeper dive on the components we identify here is best suited for subsequent research, some of which we identify in the following section.

Implications for Theory and Research

Our analyses highlight several opportunities and needs for future research. All of the situational variables examined in our meta-analysis were all indexed in the form of individuals' perceptions of environmental factors. However, situational influences that traverse levels of analysis necessitate multi-level designs (Mathieu & Chen, 2011), so future research should employ such designs to model properly situational influences on informal learning relationships (Mathieu & Tesluk, 2010). Moreover, the variability of the effect sizes that we observed here suggests the influence of contextual moderators which can best be modeled properly using multi-level designs. We also encourage future meta-analyses to use more sophisticated methods of assessing methodological rigor and publication bias, such as the approach illustrated in the [Appendix](#).

The field could also benefit from methodological advances. One critical need is the development and validation of a measure of ILBs. To date, a wide variety of measures of ILBs have been employed, with less consistency than desirable across studies. The use of a well-articulated and validated measure of ILBs would permit better comparability across studies and accumulation of knowledge. We also encourage future researchers to examine a wider array of potential antecedents and outcomes of ILBs and to employ research designs that support stronger causal inferences. For example, we did not find any studies that controlled for previous performance levels when testing ILB–outcome relationships. Future studies might also explore patterns of ILBs over time using true longitudinal designs, given the complexities of exploring constructs that may not change in a linear fashion (Ployhart & Vandenberg, 2010). Given the potential positive outcomes of ILBs, researchers should also consider how various forms of interventions might encourage or accelerate ILBs.

Finally, we encourage future investigators to consider the potential dark side of ILBs. We suspect that engaging in informal learning may be unwelcome or even counterproductive

in some circumstances. For example, employees may develop bad habits or poor work practices (e.g., taking safety “shortcuts”) if they are influenced by negative role models and misguided incentives while “learning” on the job. Engaging in ILBs might present a distraction in some instances or be perceived as unprofessional behaviors by others. Moreover, an overemphasis on informal learning might discourage engagement in critical or foundational formal training. Some jobs require employees to follow standard operating procedures without deviation, for example when consistent or high-reliability operations are needed to ensure smooth coordination with others. Where best practices have already been established, misguided experimentation and improvisation might have negative consequences, both for performance and for an employee's reputation. In short, we should not assume that engagement in ILBs will always lead to positive outcomes.

Implications for Practice

As noted, it was not within the scope of this paper to explore whether antecedents and outcomes of informal learning differ from those for formal learning. From a theoretical perspective, some might suggest it is important to know whether the antecedents and outcomes for the two learning constructs are the same. Our view, however, is that even if the antecedents and outcomes do not differ, there are still important applied implications to understanding the factors that contribute to, and stem from, informal learning. For instance, in situations where there are limitations on formal learning opportunities or where formal learning is not a feasible option altogether (e.g., due to budgetary or logistical constraints), encouraging and supporting informal learning might provide an alternate, less resource-intensive means for developing important knowledge and skills. Formal training is aimed at the development and application of specific behaviors and skills, which could have a somewhat narrow applicability. In contrast, informal learning behaviors can be used on a continuous basis to develop an unlimited and broad scope of skills, including those not targeted during formal training.

Organizations should consider conducting an “informal learning needs analysis.” While a traditional training needs analysis identifies what trainees should know or be able to do *by the end of training*, an informal learning needs analysis would focus on what individuals need to (and can) learn informally *on the job*. It could also help reveal where informal learning may be detrimental (e.g., where prescribed behaviors are required and experimentation is problematic). Communicating common informal learning needs and opportunities to employees could enable them to make constructive choices with their discretionary learning time.

Given that some individuals may be naturally more inclined or equipped to engage in informal learning behaviors

than others, it may be beneficial to help individuals become better “informal learners” by providing them with tips and insights about effective ILBs (e.g., to become better at asking questions, listening actively, recognizing patterns, soliciting feedback, or observing experts). These tips and insights could be shared in a number of ways, from one-on-one coaching sessions to formal training sessions. One might envision embedding tips or advice into an existing training program to help trainees become better equipped to engage in informal learning behavior after the training. Employees increasingly enter dynamic or unanticipated situations, so building their general informal learning skills may help them when new requirements and conditions emerge.

Given the robust links between perceived support and ILBs (although observed effect sizes in operational settings may be somewhat smaller than those reported here as we corrected for artifacts), organizations should take actions to support and encourage informal learning. They can formally communicate the value of ILBs, by highlighting potential learning opportunities. They can also teach employees to recognize “triggers” for learning opportunities. Research on the development of adaptive expertise (Smith et al., 1997) and adaptive transfer (Bell & Kozlowski, 2008) may provide additional insights for those looking to develop programs related to ILBs. Organizations might also provide informal support by encouraging managers to model, support, and recognize ILBs such as seeking feedback and experimentation. Moreover, organizations should take care to reduce obstacles to informal learning (Kerr, 1975), such as lack of time. Finally, in organizations where continual, informal learning is required, such as those in rapidly changing business environments, an attempt could be made to hire employees who are more likely to engage in ILBs. For example, assessing personality characteristics such as openness to experience during the hiring process (e.g., Noe, et al., 2013) might increase the likelihood that informal learning becomes a naturally occurring phenomenon in the organization.

Limitations

Our meta-analyses have a few constraints and limitations worth noting. Our higher-order categories for the meta-analyses were more heterogeneous than we would have liked. Whereas a wide variety of individual predispositions may motivate people to engage in ILBs, some are likely to be more valuable than others. This creates some heterogeneity of effect sizes which may be reduced with more targeted analyses of specific antecedent relationships when more studies become available. Aggregation into higher-order categories also impedes the ability to draw inferences directly at the construct level (although we do provide all primary data in Table 5 for subsequent researchers to more fully parse). Inevitably, all meta-analyses are confronted

with the tradeoff between categorizing variables to yield a sufficient number for analysis, versus the precision of those categories (Cortina, 2003).

From a methodological standpoint, there are inherent challenges in measuring ILBs because they are, by their very nature, individually driven and maintained. Because ILBs occur outside a formal course curriculum and without specified learning objectives (Eraut, 2004), it can be more difficult to link the outcomes of ILBs to specific outcomes than is the case in research on formal training interventions. Moreover, ILBs are typically measured not at an event-specific level (e.g., “Do you engage in informal learning on task X? Task Y? Task Z?”) but at a more general context-specific level (e.g., “Do you engage in informal learning at work? At home? At school?”), although we did ensure adherence to the principle of correspondence (Ajzen & Fishbein, 1977) by linking ILBs only with variables at commensurate levels of specificity.

Finally, although often bestowed with an undue air of objectivity, meta-analysis is something of an art (Cerasoli et al., 2014; Rosenthal & DiMatteo, 2001). We have followed guidelines put forth by others (Aguinis, Gottfredson & Wright, 2011; Aguinis, Pierce, Bosco, Dalton & Dalton, 2011; Cooper, 2003) to reduce between-coder error and subjectivity, which includes holding multiple discussions, maintaining extensive documentation, examining cross-coder agreement, and publishing of primary data. Nevertheless, we should acknowledge that most of the data are cross-sectional in nature, so causal claims should be made with care. For example, there are likely some reciprocal relationships between rank/tenure and ILBs: engagement in ILBs breeds better performance, which increases rank/tenure, thereby reinforcing ILBs.

Conclusion

Despite the acknowledgment that a great deal of organizational learning occurs outside of formal training, companies have devoted relatively little attention and few resources towards discovering ways to foster IL in working adults (Berg & Chyung, 2008). Given the effects observed in the current study, we echo the sentiments expressed by Marsick et al. (1999) that “...informal learning from experience cannot be left completely to chance...” (pp. 93–94). Fortunately, we also identified several factors that organizations can leverage to promote and institutionalize informal learning. To support this organizational need, research should continue to explore and clarify the antecedents and consequences of ILBs. This review and meta-analysis helped highlight what is currently known about ILBs and raised some additional questions. We encourage researchers to explore these questions in future research.

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Appendix: Methodological Rigor Analyses

Addressing the issue of primary data publication quality of the studies included in our meta-analysis is critical in this case. Given the ubiquity with which informal learning occurs, we intentionally conducted a very broad search and did not set any a priori exclusionary criteria with respect to the source of the primary data. This of course introduces the possibility that research from arguably less rigorous outlets may impact the observed effect size of a given study.

We explored this issue in depth and conferred with several of the leading scholars on meta-analysis. The leading scholars suggested that, perhaps, we could rate the quality of each study and correlate those ratings with the reported effects sizes. However, they were quick to note that ratings of study quality are notoriously unreliable (e.g., Armijo-Olivo et al., 2012) and that may not be the best course of action. So we decided to be more systematic and rigorous about testing for such effects using a two-pronged approach. First, we indexed the quality of the outlet where each sample appeared using study citations and journal impact factors. Second, we coded whether informal learning–correlates were concurrent versus lagged in time and whether they were measured using the same or different methods.

In terms of indexing the quality of each study, we obtained its number of citations in Google Scholar as of October 2016. These ranged from 0 to 424 (mean = 51.33, SD = 94.35). Second, we obtained the journal impact factor scores for all sources except dissertations and theses, which ranged from .10 to 4.80 (mean = 1.15, SD = 1.09). For analytic purposes, we assigned an impact value of zero to uncited dissertations and theses yielding a distribution mean of .82 (SD = 1.06). We used Google Scholar because it tracks citations more liberally for all sorts of publications than do the more restrictive citation bases. Whereas this may yield larger citation rates, such inflation should be consistent across the studies sampled here and therefore not bias these analyses. As one would anticipate, source impact factor scores and citations correlated significantly ($r = .70, p < .001$). Naturally older studies have a higher likelihood of being cited (year published correlated $r = -.54, p < .001$ with citations). Nevertheless, controlling for year published, source impact scores still correlated significantly with citations ($r = .67, p < .001$). Therefore, we believe that we have two widely used indices of source quality that correlate as expected.

Of the 376 effects sizes that are included across the various meta-analyses, 90% were concurrent, and 10% were lagged or from an experiment. For analytic purposes, we

coded concurrent measures = 0 and lagged/experimental = 1. As for measurement source, both informal learning and the correlate were measured using the same source (coded = 1) 98% of the time and 2% were from differing sources (coded = 0). We should note that some correlations reported in a given study may have been from the same time, whereas others may have been lagged. Similarly, some correlations may have been between variables measured the same way, whereas others may have involved multiple measures. Therefore, these design features can co-vary at the effect size level of analysis.

The lower left triangle of Table 6 contains correlations among the study effect sizes, design features, and source quality indices ($N = 376$). In terms of these correlations, it is notable that the only significant ones with the observed effect sizes is a positive one for citation rate ($r = .12, p < .05$) and a negative one for sample size ($r = -.20, p < .001$). The upper right triangle of Table 6 shows the parallel partial correlations controlling for both publication year and sample size. Still, the only significant correlation with effect size is a positive one with citation rate ($r = .13, p < .05$).

The 376 effects sizes that we considered came from 49 different samples that were clustered into eight different categories for meta-analytic purposes. In other words, each sample contributed, on average, 7.7 effects sizes to the meta-analyses. This lack of independence needs to be taken into account in any pooled analysis associating the magnitude of effect sizes with study characteristics. In addition, the different meta-analysis categories revealed significant mean differences in effects sizes that should also be accounted for in any such analysis. Accordingly, we do so using a cross-classified hierarchical linear model analysis of effects sizes as described below (HLM; Raudenbush, Byrk, Congdon, & du Toit, 2004). Cross-classified models simultaneously account for multiple level 2 nesting arrangements, which in this case

Table 6 Correlations between study design feature, source quality, and observed effect sizes

Variables	1	2	3	4	5	6
1. Effect size		.01	-.05	.13	.08	
2. Design ^a	.04	–	-.42	.12	.36	
3. Measurement ^b	-.05	-.41	–	-.39	-.34	
4. Citations	.12	.10	-.41	–	.76	
5. Journal impact	.10	.37	-.34	.73	–	
6. Year	.03	.08	.12	.29	.02	–
7. Sample size	-.20	-.11	.03	-.09	-.15	.11

$N = 376$ effect sizes from 49 samples. Given the lack of independence of observations, significance levels are biased and should be interpreted cautiously. Correlations: $|.10|, p < .05$; $|.14|, p < .01$

^a Coded: concurrent = 1, lagged/experimental = 2

^b Coded: same source = 1, different source = 0

map to samples and meta-analysis categories. Although not precise estimates in the context of multi-level models, we report overall effect sizes for these models (i.e., $\sim R^2$) using the formulas advanced by Snijders & Bosker (1999).

The HLM model had 376 level 1 effects sizes that were cross-classified in terms of 49 level 2 samples and 8 level 2 meta-analysis categories. A baseline (i.e., null model) revealed that 30% of total effects size variance resided across samples [$\chi^2(49) = 201.39, p < .001$], 17% fell across meta-analysis categories [$\chi^2(7) = 119.87, p < .001$], and 53% remained unaccounted for. Therefore, clearly it is importantly to account for both samples and categories. Notably, given that meta-analyses adjusts for the influence of differing sample sizes on observed effects sizes, we introduced sample size as a weighting factor at level 1 in the remaining HLM analyses. Sample size alone accounted for 71% of the total observed variance in effects sizes, which is consistent with the pattern observed across the individual meta-analyses. Notably, however, even after accounting for sample size influences, significance between sample [$\chi^2(48) = 109.15, p < .001$], between meta-analysis categories [$\chi^2(7) = 324.13, p < .001$], and variance in effects sizes remained.

We next introduced the design and measurement dummy codes as level 1 predictors. Notably, we centered-within-cluster (CWC) these scores per meta-analysis category so as to consider their influence per meta-analysis (see Enders & Tofighi, 2007). Together the two dummy codes account for a nonsignificant [$\Delta\chi^2(2) = 2.16, ns$] 1% $\Delta \sim R^2$ and neither exhibited a significant unique influence. We then introduced the average category design and measurement features as potential level 2 predictors. The logic here is that some meta-analysis categories may be more likely to contain certain design features than others. For example, 100% of the informal learning–attitudes correlations were concurrent, whereas 49% of the informal learning–individual predisposition correlations came from lagged designs. Nevertheless, introducing the level 2 category average design features also failed to account for any significant additional variance [$\Delta\chi^2(2) = 2.31, ns$, $\Delta \sim R^2 = 2\%$] and neither mean value exhibited a significant unique influence. These findings provide clear evidence that the magnitude of effects sizes was not significantly related to design or measurement source factors, either considered within or between meta-analysis categories.

Finally, we introduced year of publication, citations, and journal impact factors as level 2 predictors of effects sizes from the sample classification function (retaining the above effects in the equation). Collectively, adding them to the equation accounted for a significance [$\Delta\chi^2(3) = 13.03, p < .05$] $\Delta \sim R^2 = 2\%$. However, the only significant unique effect was, oddly, attributable to a negative effect associated with the year of publication control variable [$T = -.005, SE = .002, p < .05$].

The summary results of this analysis are shown below in Table 7. In sum, after accounting for the double nesting and

Table 7 Cross-classification analysis of effects size estimates as related to design features and source quality

Predictors	Parameter Estimate	Standard Error	<i>t</i> value
Within predictors			
Design ^a	.060	.096	.62
Measurement ^b	.178	.149	1.20
Between category			
Mean design ^a	.216	.219	.99
Mean measurement ^b	-.503	.502	-.97
Between sample			
Publication year	-.005	.002	-2.34*
Citations	.0004	.0002	1.51
Source impact	-.021	.022	-.97

Three hundred seventy-six effect sizes in 49 samples crossed with 8 meta-analysis categories

* $p < .05$

^a Coded: concurrent = 1, lagged/experimental = 2

^b Coded: same source = 1, different source = 0

varying sample sizes, neither design nor measurement features were related significantly to the magnitude of effect sizes, considered either centered within meta-analysis category or on average across categories. Source quality, indexed both in terms of citations and impact scores (controlling for year of publication), also had no significant relationships with the effects sizes.

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